Technical Specification & Architecture Blueprint

Abundance Solutions Offline-First Hybrid Platform (Admin & E-Commerce PWA)

Target Market: Zimbabwe (Low-Data / WhatsApp Bundle Optimized)

Tech Stack: Vite + Vercel + IndexedDB + Turso/Supabase

1. Executive Summary & Core Objectives

This technical specification outlines the software architecture for building a robust, high-performance Progressive Web Application (PWA) for **Abundance Solutions**. Designed to overcome local infrastructure hurdles—such as intermittent connectivity, expensive mobile data tariffs, and heavy reliance on WhatsApp data bundles—the platform serves two distinct user personas:

- **Commercial Prospect (Frontend & E-Commerce):** A lightning-fast public showroom displaying heavy industrial assets (generators, excavators, concrete mixers, tools) optimized for low-bandwidth browsing and instant WhatsApp cart conversion.
- **Business Owner & Field Admin Dashboard:** A secure, offline-capable operations suite for managing real-time field inventory, tracking cash/EcoCash financial logs, and queueing background data synchronization.

2. Technology Stack Architecture

Layer	Selected Technology	Architectural Purpose
Frontend Build Tool	Vite + `vite-plugin-pwa`	Fast Hot Module Replacement (HMR), static asset compilation, and Workbox-powered service worker caching.
Hosting & Edge	Vercel Serverless & Edge Network	Global content delivery, SPA routing rewrites, and serverless API execution for cloud sync webhooks.
Client-Side Storage	Dexie.js (IndexedDB wrapper)	High-capacity local database holding products, transaction outbox, financial logs, and offline state.
Cloud Database	Turso (SQLite at Edge) or Supabase	Relational cloud persistence supporting transactional sync and Row Level Security (RLS).
Conversion Bridge	WhatsApp Click-to-Chat API	Bypasses costly online mobile data payment gateways by formatting and transmitting orders via WhatsApp bundles.

3. Offline-First Storage & Smart Resync Engine

To ensure zero data loss during network disruptions, mutations are written locally first before attempting cloud propagation.

3.1 The Local Outbox Queue Pattern

When an admin user logs a financial entry or updates field inventory offline, the application executes the following routine:

1. **Immediate Local Write:** Record is saved to IndexedDB with metadata *syncStatus = 'pending'* and a unique UUIDv4.
2. **Optimistic UI Render:** The dashboard updates instantly without blocking user workflow.
3. **Network Event Listener:** The app monitors connectivity via *window.addEventListener('online', triggerSync)*.
4. **Batch Reconcile:** Upon restoration, pending outbox payloads are flushed to Vercel API endpoints. Once confirmed by the cloud database, local status updates to *syncStatus = 'synced'*.

> **Conflict Resolution Strategy:** Uses a *Last-Write-Wins (LWW)* timestamp vector combined with server-side audit logs to prevent race conditions during multi-device warehouse updates.

4. Zimbabwe-Centric Optimization: The WhatsApp Bundle Bridge

Recognizing that commercial prospects frequently operate under restrictive WhatsApp-only data bundles where open web socket connections fail, the e-commerce engine features an offline-to-social conversion bridge.

4.1 Cart Serialization Flow

- Users browse cached inventory catalogs (generators, JCB units, grinders) completely offline.
- Items are added to a local shopping cart state managed via IndexedDB.
- Upon clicking "Checkout via WhatsApp", the system formats raw cart data into a pre-encoded URL string:

URL = https://wa.me/263719450765?text=OrderSummaryPayload

This triggers the native WhatsApp application on mobile or desktop, allowing the transaction to initiate successfully without consuming expensive general internet data bundles.

5. Performance & Data-Saving Protocols

To adhere to strict bandwidth limits, asset delivery is tightly regulated:

- **Incremental Delta Updates:** Utilizing Workbox manifest strategies to download only altered JSON data blocks rather than re-fetching full application bundles.
- **Compressed Media Assets:** Heavy machinery photography is served in next-gen WebP format with responsive multi-resolution srcset attributes.
- **Bandwidth Saver Toggle:** Allows users in low-signal or high-tariff zones to disable high-definition image prefetching entirely in favor of lightweight text-and-thumbnail summaries.